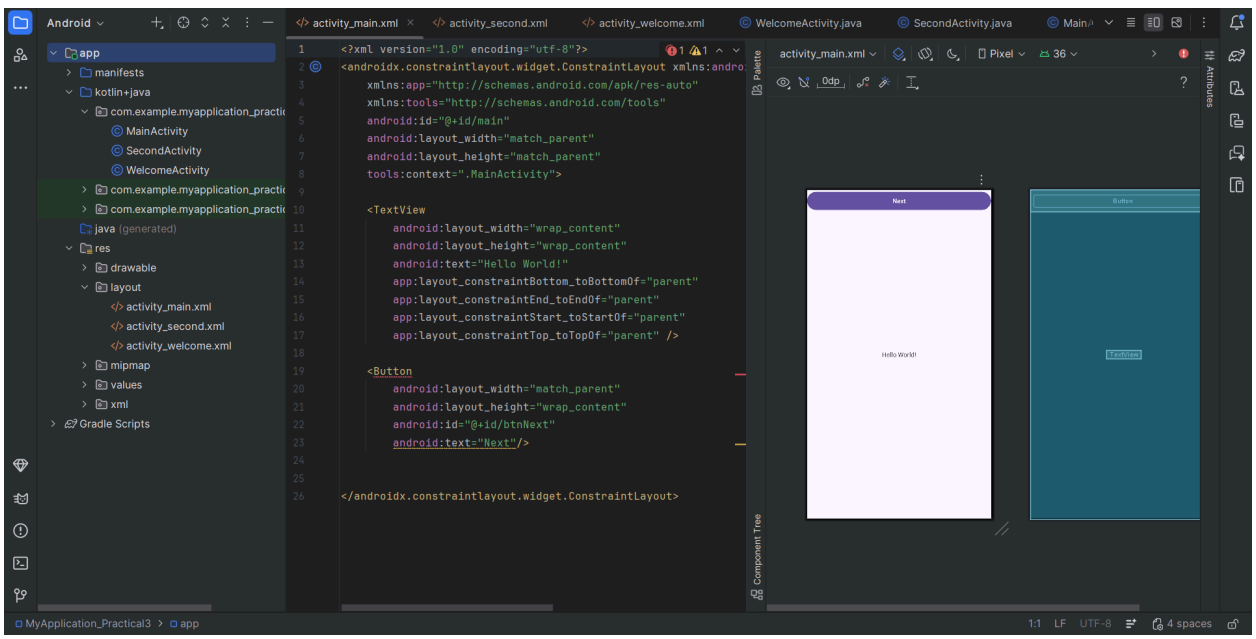
Practical 3

Aim: Understanding Android Components. Create an Android application with multiple activities. Implement the life-cycle methods for each activity and observe their execution.

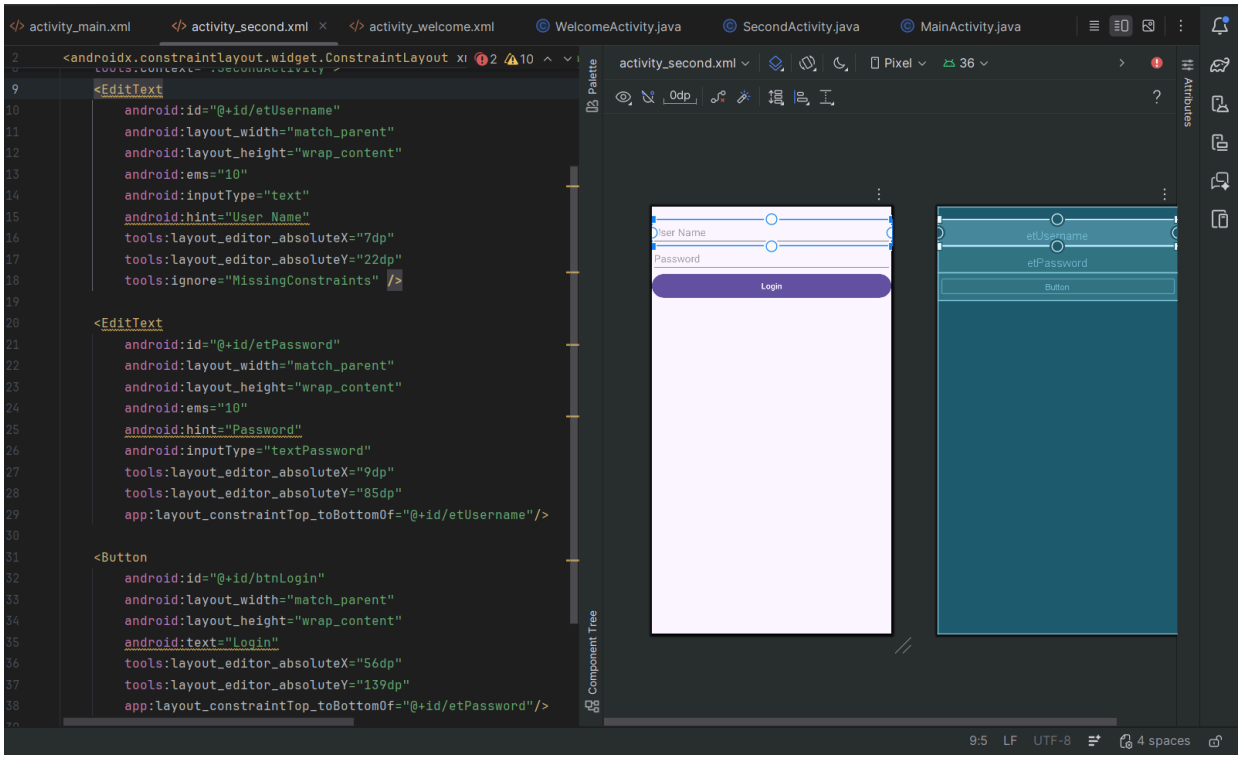
Theory: This practical demonstrates Android activities, where each activity represents a screen, shows navigation using Intent, demonstrates data passing between activities, performs basic input validation, and uses onCreate() in the activity lifecycle. The MainActivity opens SecondActivity, SecondActivity takes the username and password, and WelcomeActivity displays the welcome message.

Codes:

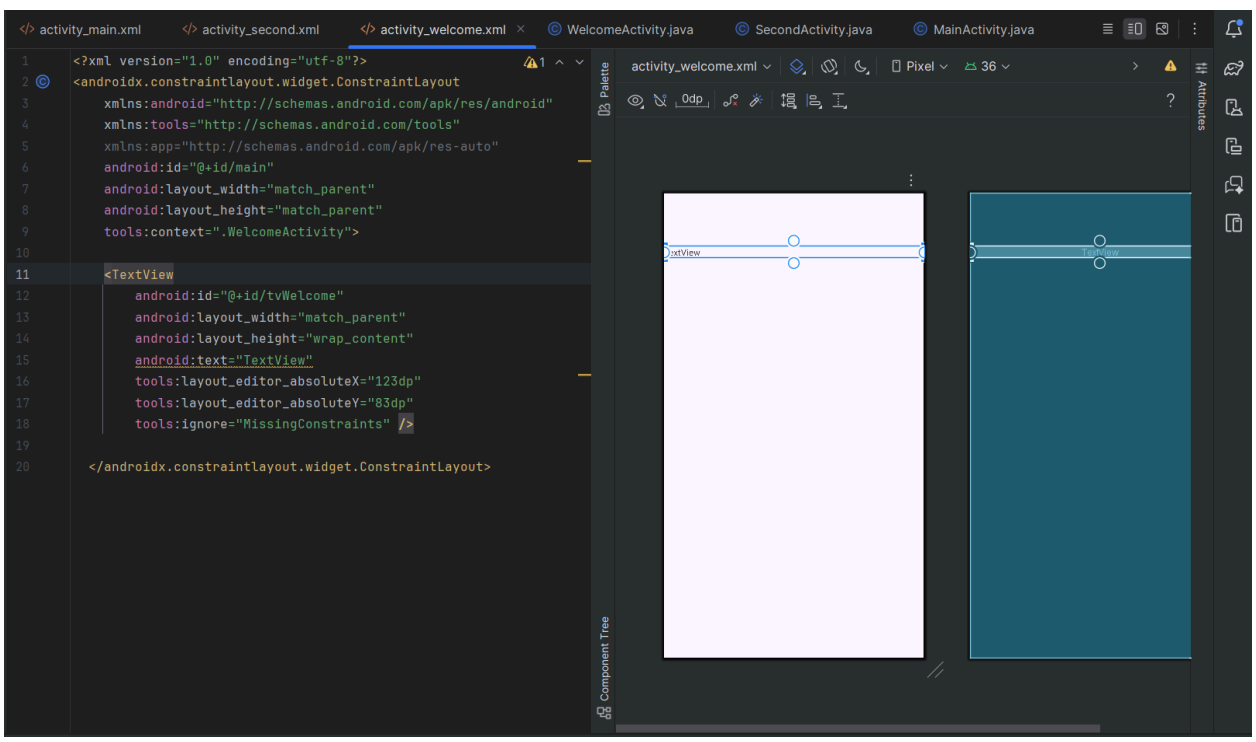
activity_main.xml - Defines the app's first screen with a "Next" button that navigates the user to the second activity.



activity_second.xml - Creates a login interface with username and password fields, along with a Login button to proceed further.

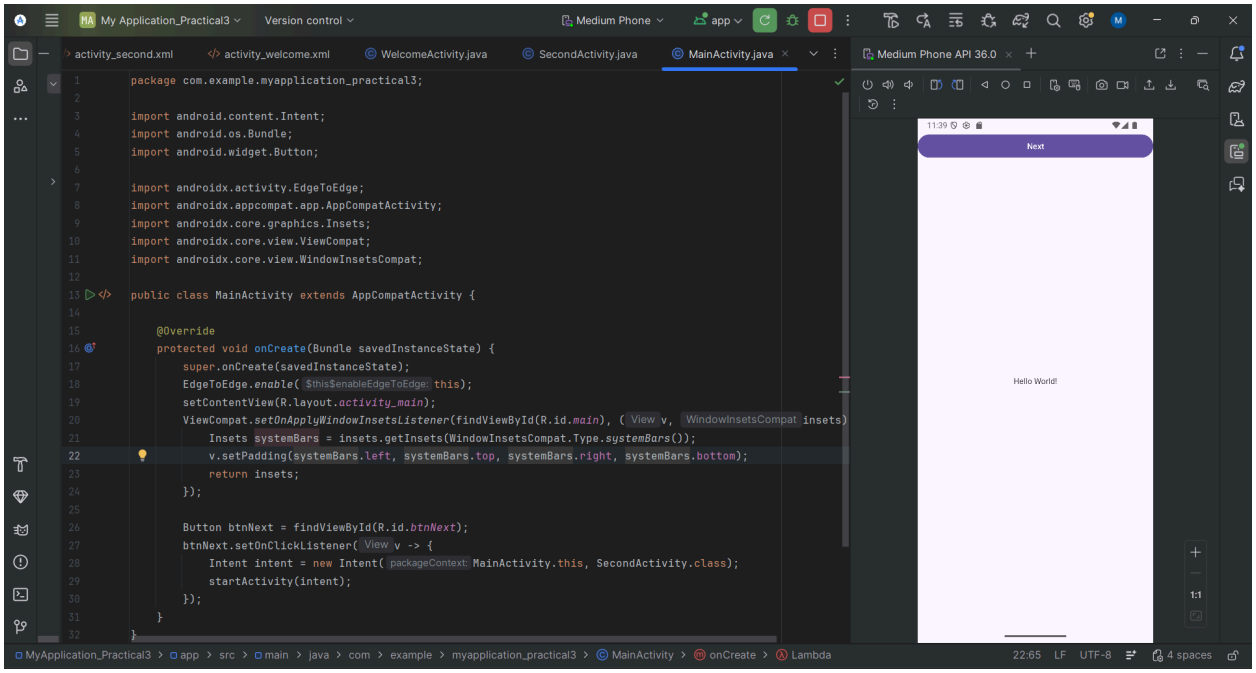


activity_welcome.xml - Displays a welcome message using TextView, showing the username passed from the previous activity.



Output:

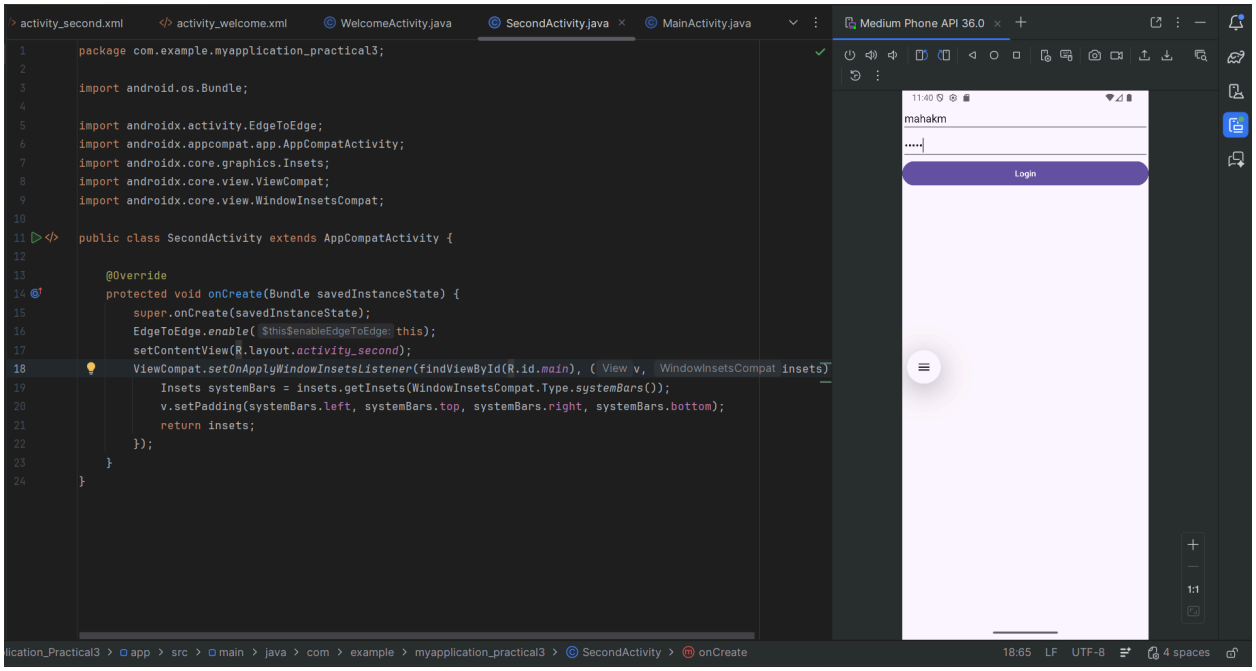
MainActivity.java - This is the application's first screen. It contains a button that navigates the user to the SecondActivity when clicked.



Code Explanation

- onCreate() → Initializes the activity and loads layout
- Button btnNext → Used to move to next screen
- setOnClickListener() → Handles button click
- Intent → Used to open SecondActivity
- startActivity() → Starts new activity

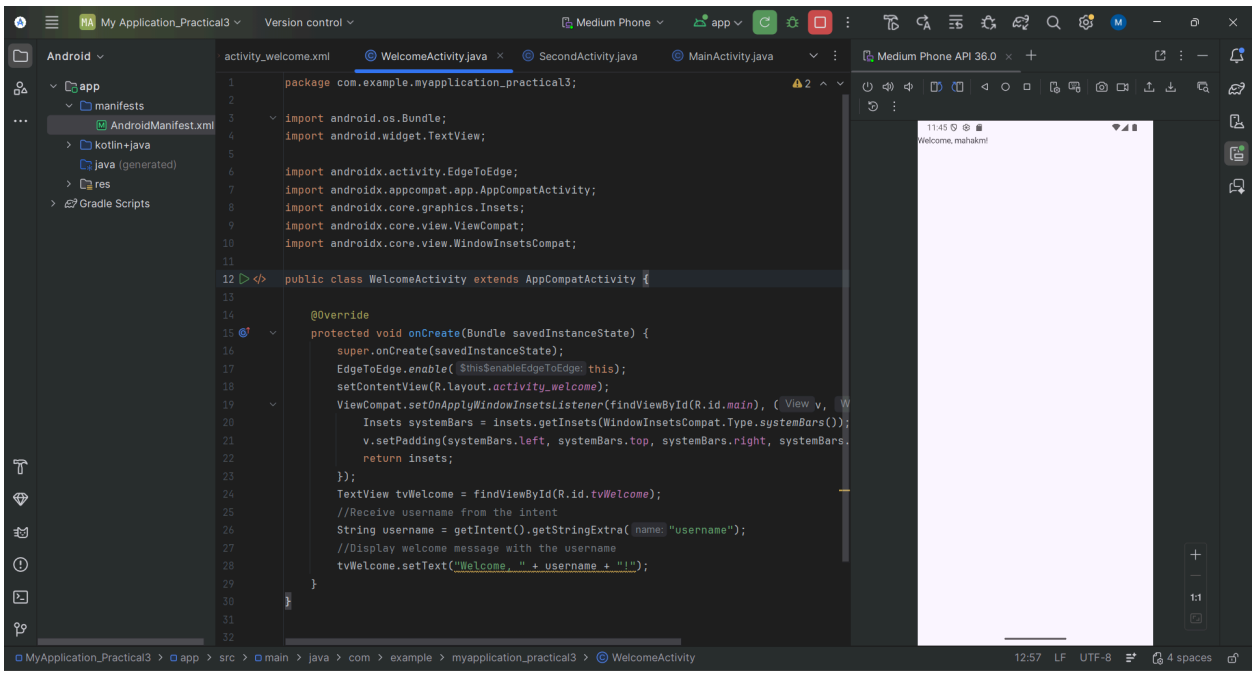
SecondActivity.java - This activity acts as a login screen where the user enters username and password. It validates input and sends data to WelcomeActivity.



Code Explanation

- EditText → Takes username and password
- Button btnLogin → Performs login action
- setOnClickListener() → Handles click event
- if (username.isEmpty()) → Validates input
- Toast → Displays error message
- Intent.putExtra() → Sends username
- startActivity() → Opens WelcomeActivity

WelcomeActivity.java - This activity displays a welcome message using the username received from SecondActivity.



Code Explanation

- onCreate() → Initializes activity
- getIntent().getStringExtra() → Receives username
- TextView → Displays message
- setText() → Shows “Welcome, username!”

Conclusion

The Android application with multiple activities was successfully developed. The activity lifecycle methods were implemented and observed using Logcat. Navigation between activities and data passing using Intent was also successfully demonstrated.